

Purpose/Objective(s): To evaluate long-term outcomes after surgery and radiation therapy (RT) for spinal myxopapillary ependymoma (MPE).

Materials/Methods: We reviewed 51 patients with spinal MPE treated in 1968-2007. Median follow-up time was 11 years (range, 0.2-37 years). Overall survival (OS), progression-free survival (PFS), and local control (LC) rates were calculated by Kaplan-Meier estimates. Clinical and treatment variables (age at diagnosis, sex, tumor location, symptom lateralization, initial neurologic function, primary treatment modality, extent of surgical resection) were analyzed for prognostic value with log-rank tests and Cox regression analysis. Statistical analysis was done with SPSS. All tests were two-sided, and $P \leq 0.05$ was considered statistically significant.

Results: Median age at diagnosis was 35 years (range, 8-63 years); 32 patients were male and 19 female. Tumors were in the thoracolumbar cord in 14 patients (27%) and lumbosacral cord/cauda equina in 37 patients (73%). Primary treatment was surgery in 20 patients (39%), surgery plus RT in 30 patients (59%), and RT alone in 1 patient (2%). Of the 50 patients who had had surgery, 28 (56%) had gross total resection, 18 (36%) subtotal resection, and 1 (2%) biopsy only. Ten-year OS, PFS, and LC rates for the entire group were 93.2%, 63.9%, and 65.2%, respectively. Seventeen patients (33%) had disease recurrence, all in the neural axis: 13 at the primary site only, 1 at distant neural axis only, and 3 at both. On univariate analysis, age > 35 years at diagnosis and use of adjuvant RT were associated with improved PFS ($P = 0.018$ and $P = 0.044$) and LC ($P = 0.007$ and $P = 0.020$). On multivariate analysis, age at diagnosis (> 35 years) and adjuvant RT remained significant predictors of PFS (adjusted hazard ratios [HRs] = 0.337 for age, 95% CI 0.148-0.766, $P = 0.009$, and 0.403 for RT, 95% CI 0.191-0.848, $P = 0.017$), and LC (adjusted HRs = 0.148 for age, 95% CI 0.040-0.544, $P = 0.04$; and 0.437 for RT, 95% CI 0.250-0.762, $P = 0.04$). Among the 28 patients with gross total resection, local recurrence rates were 37.5% (6/16) with resection alone vs. 16.7% (2/12) for those who were treated with adjuvant RT ($P = 0.10$). None of the variables tested influenced OS.

Conclusions: Patients with MPE typically have long-term survival. Disease recurrence occurs primarily at the primary sites. Adjuvant RT is associated with reduced local recurrence and improved PFS.

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Feasibility of Radiosurgical Decompression of Metastatic Epidural Spinal Cord Compression (MESCC) in Nonoperable Patients

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Purpose/Objective(s): To prospectively test the hypothesis that spine stereotactic radiosurgery (SSRS) used as the primary treatment of inoperable MESCC is both feasible and safe.

Materials/Methods: Two ongoing phase I/II trials enrolling patients > 18 years old, with MESCC on spine MRI (ranging from involvement of epidural space not compressing the dura, to compression of the spinal cord obliterating all visible CSF at that level) and maximum of 2 contiguous vertebral levels involved with metastasis to be irradiated in a single session, motor strength $\geq 4/5$, deemed to be inoperable (patient refusal, by neurosurgical evaluation, or for any reason). In the previously irradiated trial patients are limited to one prior course of spine radiation therapy < 45 Gy conventional fractionation or similar biologically effective dose to the current MESCC > 3 months previously. In both trials, patients have a goal GTV dose of 18 Gy in a single fraction with goal CTV dose of 16 Gy. Spinal cord constraints were initially 8 Gy to 0.01 cc in the previously irradiated trial and 10 Gy to 0.01 cc in the radiation naïve trial. Spinal cord dose constraints are relaxed in 2 Gy increments per event (recurrence/progression) in each protocol while maintaining the same goal target doses in a prospective manner.

Results: Current accrual to the previously irradiated protocol is 5 patients, and radiation naïve protocol 7 patients. For these 12 patients, consult to simulation time was 0-19 days (mean 5.25 days, S.D. 5.86 days). Simulation to treatment time was never longer than 12 days (mean 7.08 days, S.D. 3.37 days). Time from consult to treatment for all patients was 6-23 days (mean 12.33 days, S.D. 5.71 days). Actual planning times were documented. Utilizing class solutions, time from contour completion to first plan review averaged 8.18 hours (2.25-16 hours, S.D. 4.48 hours). Initial plan review to plan approval ranged from 0-6 hours (mean 2.42 hours, S.D. 2.31 hours). Overall, planning start to plan approval averaged 10.6 hours (2.3-18 hours, S.D. 4.83 hours). With a current mean follow up of 5.07 months (0.59-10.59), four of 12 patients have recurred/progressed, and four have died. Mean time to recurrence was 6.42 months (1.41-8.98). At present, no radiation myelopathy has been detected.

Conclusions: The prospective trials are ongoing but this pilot data shows feasibility of SSRS in the treatment of inoperable MESCC, and that it can be performed in a timely fashion with spine class solutions. Future data to be reported on control, toxicity, patterns of failure and maximum tolerated dose defined for spinal cord.

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Long-term Result of Carbon-ion Radiation Therapy for Locally Advanced or Unfavorably Located Choroidal Melanoma: CT Based 2-port Orthogonal Therapy Can Reduce the Incidence of Neovascular Glaucoma

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Purpose/Objective(s): To evaluate long-term result of carbon ion radiation therapy (C-ion RT) for the patients with choroidal melanoma and to determine the prognostic factors and risk factors of neovascular glaucoma (NVG).

Materials/Methods: Between January 2001 and February 2012, 116 patients with locally advanced or unfavorably located choroidal melanoma received the C-ion RT with CT based treatment planning. Of the 116 patients, 114 patients followed longer than 6 months were analyzed for this study. Of those, 106 patients had T3 tumor and 8 patients had T2 tumor according to UICC 5th edition. Initially the treatment was performed according to a phase I/II dose-escalation protocol, therefore 16 patients treated with 60 GyE, 78 with 70 GyE, 13 with 77 GyE, and 7 with 85 GyE, in 5 fractions. Since October 2005, two-port orthogonal therapy has been started in attempting to reduce the risk of NVG, and 51 patients were already performed. Dose-volume histogram analysis was additionally performed in 106 patients with regard to NVG occurrence.

Results: Median follow up duration was 4.6 years (range, 0.5-10.6). Five-year overall survival (OS), cause-specific survival (CSS), local control rate, distant metastasis free survival rate (DMFSR), and eye retention rate of entire group were 80.4%, 82.2%, 92.8%, 72.1%, and 92.8%, respectively. NVG was occurred in 36 patients, and enucleation was undergone in 3 patients because of severe glaucoma. Three-year and 5-year NVG incidence rate was 29.7% and 35.9% for all the analyzed patients. Concerning number of port, NVG incidence rate of two-port group was significantly lower than that of one-port group on univariate analysis ($p < 0.001$ log-rank test): 3-year NVG incidence rate was 41.6% and 13.9% for one-port group and two-port group. Multivariate analysis identified tumor size, distance from optic disc (0-3mm vs others), and one-port therapy were independent significant risk factors of NVG; distance from optic disc (0 vs others) was independent significant prognostic factor of OS, CSS, and DMFSR. Dose-volume histogram analysis showed the average irradiated volumes of iris ciliary body were significantly lower in non-NVG group at all dose levels than NVG group.